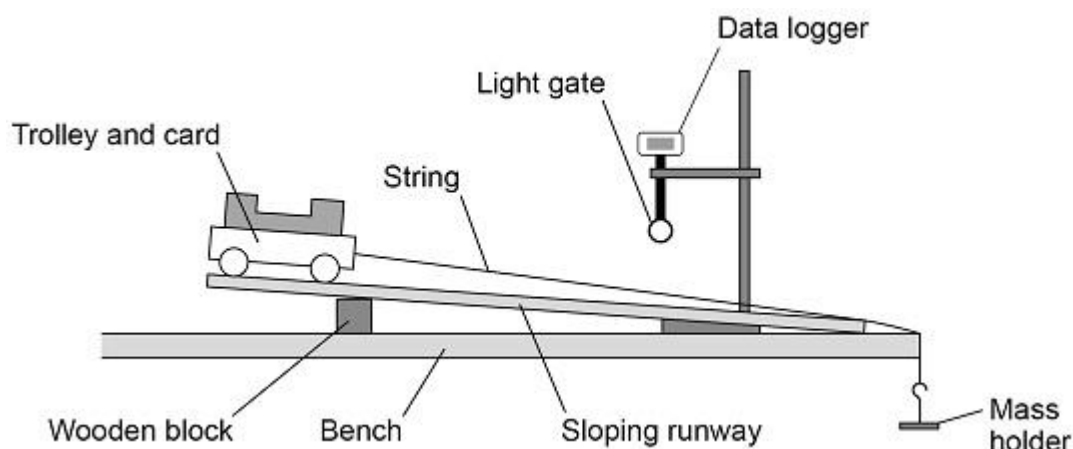


### Q1.

A student investigated the acceleration of a trolley.

The diagram below shows how the student set up the apparatus.



- (a) Before attaching the mass holder the student placed the trolley at the top of the runway. The trolley rolled down the runway without being pushed.

What change to the apparatus in the diagram could be made to prevent the trolley from starting to roll down the runway?

Tick (✓) **one** box.

Move the wooden block to the left.

☐

Shorten the length of the runway.

☐

Use a taller wooden block.

☐

(1)

- (b) The student attached the mass holder to the string.

The string rubbed along the edge of the bench as the mass holder fell to the floor.

Suggest what the student could do to prevent the string from rubbing.

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(1)

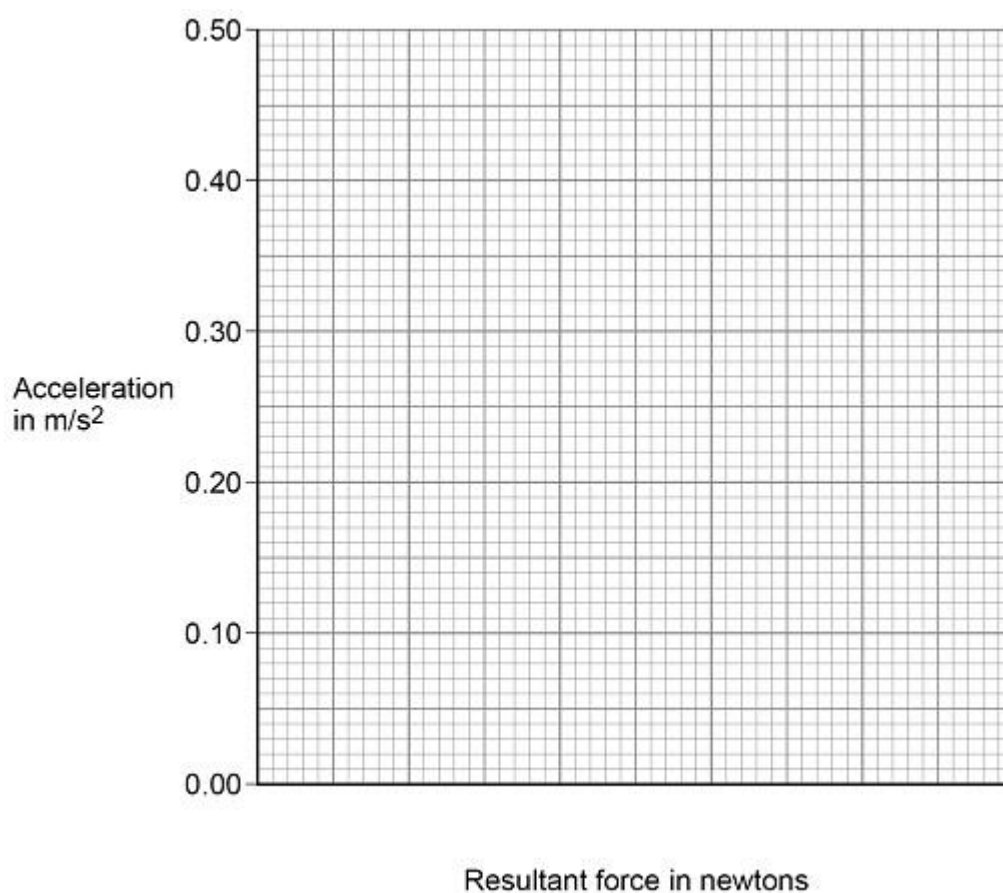
The light gate and data logger were used to determine the acceleration of the trolley.

The student increased the resultant force on the trolley and recorded the acceleration of the trolley.

The table below shows the results.

| Resultant force in newtons | Acceleration in $\text{m/s}^2$ |
|----------------------------|--------------------------------|
| 0.05                       | 0.08                           |
| 0.10                       | 0.18                           |
| 0.15                       | 0.25                           |
| 0.20                       | 0.32                           |
| 0.25                       | 0.41                           |

The graph below is an incomplete graph of the results.



(c) Complete the graph.

- Choose a suitable scale for the x-axis.
- Plot the results.
- Draw a line of best fit.

(4)

(d) Describe the relationship between the resultant force on the trolley and the acceleration of the trolley.

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(1)

- (e) Describe how the investigation could be improved to reduce the effect of random errors.

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(2)

- (f) Write down the equation that links acceleration ( $a$ ), mass ( $m$ ) and resultant force ( $F$ ).

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(1)

- (g) The resultant force on the trolley was 0.375 N.

The mass of the trolley was 0.60 kg.

Calculate the acceleration of the trolley.

Give your answer to 2 significant figures.

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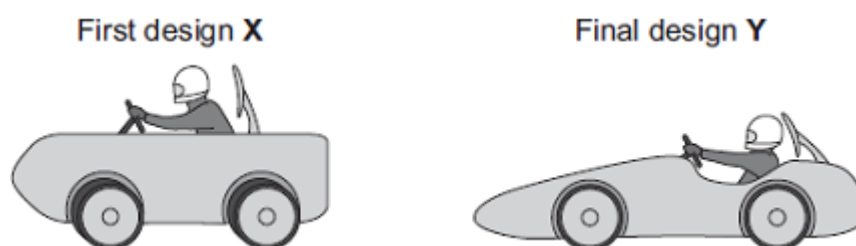
Acceleration (2 significant figures) = \_\_\_\_\_ m/s<sup>2</sup>

(4)

(Total 14 marks)

## Q2.

- (a) Some students have designed and built an electric-powered go-kart. After testing, the students decided to make changes to the design of their go-kart.



The go-kart always had the same mass and used the same motor.

The change in shape from the first design (X) to the final design (Y) will affect the

top speed of the go-kart.

Explain why.

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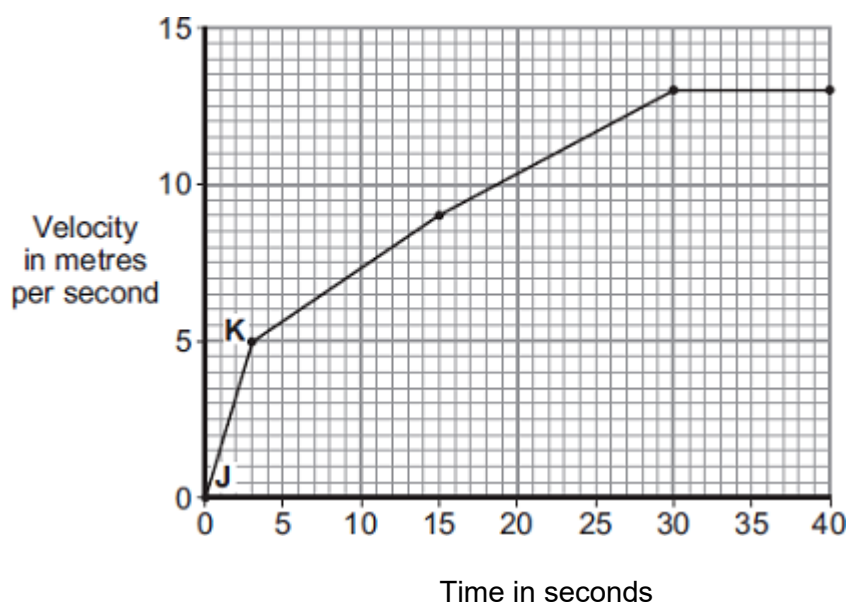
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(3)

- (b) The final design go-kart, **Y**, is entered into a race.

The graph shows how the velocity of the go-kart changes during the first 40 seconds of the race.



- (i) Use the graph to calculate the acceleration of the go-kart between points **J** and **K**.

Give your answer to **two** significant figures.

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Acceleration = \_\_\_\_\_ m/s<sup>2</sup>

(2)

- (ii) Use the graph to calculate the distance the go-kart travels between points **J** and **K**.

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Distance = \_\_\_\_\_ m

(2)

- (iii) What causes most of the resistive forces acting on the go-kart?

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(1)

(Total 8 marks)

**Q3.**

The diagram shows a shuttlecock that is used for playing badminton.



The shuttlecock weighs very little.

When you drop it from a height of a few metres, it accelerates at first but soon reaches a steady speed.

Explain, as fully as you can:

- (a) why the shuttlecock accelerates at first,

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(2)

- (b) why the shuttlecock reaches a steady speed.

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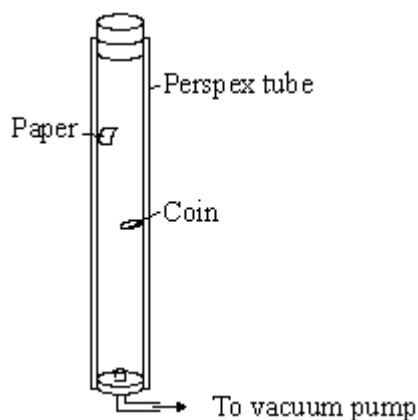
(3)

(Total 5 marks)

**Q4.**

The apparatus shown is used to compare the motion of a coin with the motion of a piece

of paper as they both fall.



- (a) When the tube is filled with air the coin falls faster than the piece of paper. Why?

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(1)

- (b) The air in the tube is removed by the vacuum pump. The tube is turned upside down.

State **two** ways in which the motion of the coin and piece of paper will change compared to when there was air in the tube.

1. 

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2. 

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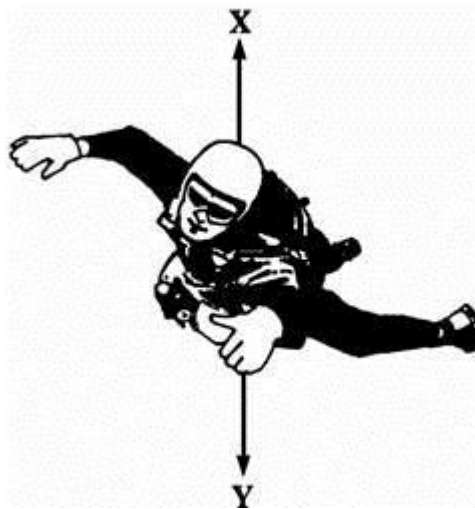
(2)

(Total 3 marks)

### Q5.

A sky-diver jumps from a plane.

The sky-diver is shown in the diagram below.



(a) Arrows **X** and **Y** show two forces acting on the sky-diver as he falls.

(i) Name the forces **X** and **Y**.

**X** \_\_\_\_\_

**Y** \_\_\_\_\_

(2)

(ii) Explain why force **X** acts in an upward direction.

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(1)

(iii) At first forces **X** and **Y** are unbalanced.

Which of the forces will be bigger? \_\_\_\_\_

(1)

(iv) How does this unbalanced force affect the sky-diver?

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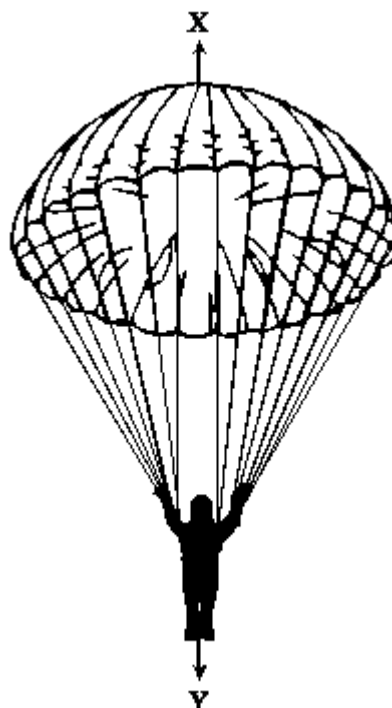


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(2)

(b) After some time the sky-diver pulls the rip cord and the parachute opens.

The sky-diver and parachute are shown in the diagram below.



After a while forces **X** and **Y** are balanced.

Underline the correct answer in each line below.

Force **X** has

*increased / stayed the same / decreased.*

Force **Y** has

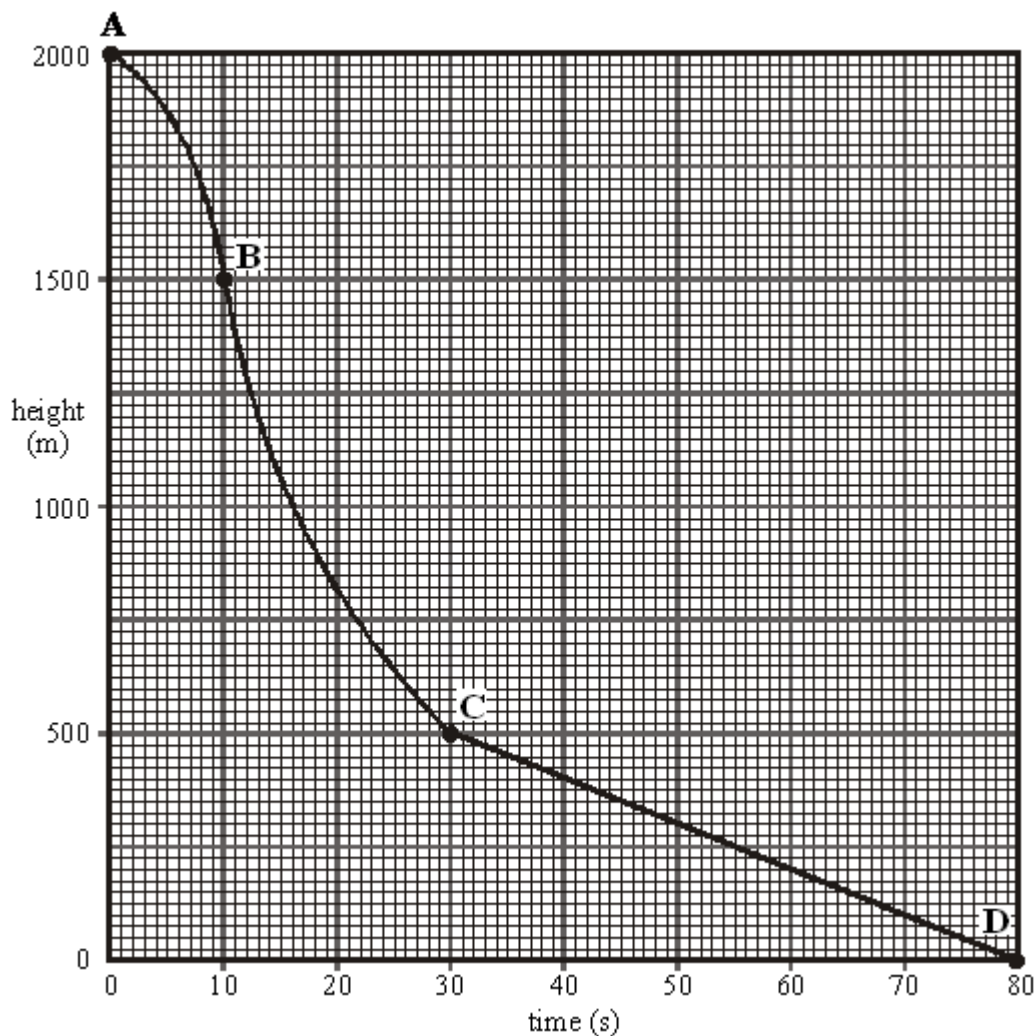
*increased / stayed the same / decreased.*

The speed of the sky-diver will

*increase / stay the same / decrease.*

(3)

- (c) The graph below shows how the height of the sky-diver changes with time.



- (i) Which part of the graph, **AB**, **BC** or **CD** shows the sky-diver falling at a constant speed?

\_\_\_\_\_ (1)

- (ii) What distance does the sky-diver fall at a constant speed?

Distance \_\_\_\_\_ m (1)

- (iii) How long does he fall at this speed?

Time \_\_\_\_\_ s (1)

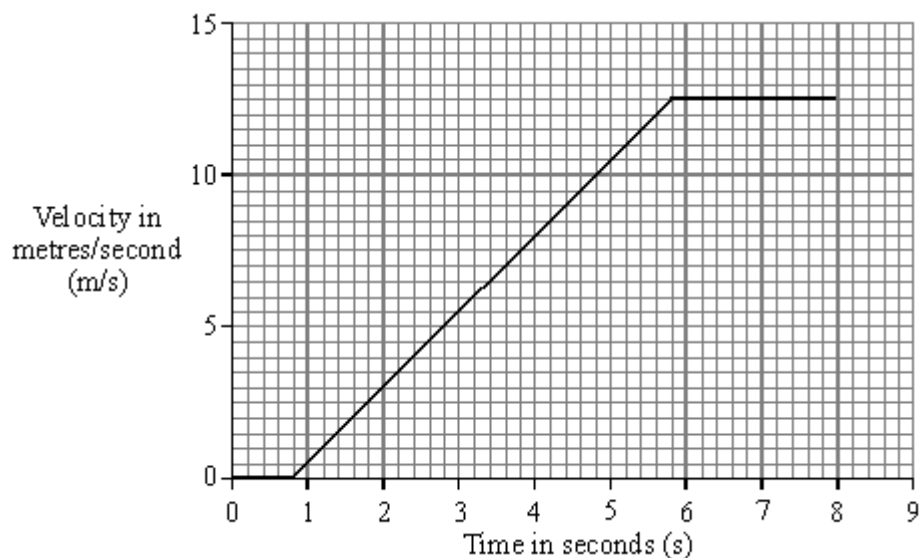
- (iv) Calculate this speed.

\_\_\_\_\_  
 \_\_\_\_\_  
 \_\_\_\_\_

Speed \_\_\_\_\_ m/s

**Q6.**

A car travelling along a straight road has to stop and wait at red traffic lights. The graph shows how the velocity of the car changes after the traffic lights turn green.



- (a) Between the traffic lights changing to green and the car starting to move there is a time delay. This is called the reaction time. Write down **one** factor that could affect the driver's reaction time.

\_\_\_\_\_ (1)

- (b) Calculate the distance the car travels while accelerating. Show clearly how you work out your answer.

\_\_\_\_\_  
\_\_\_\_\_

Distance = \_\_\_\_\_ metres (3)

- (c) Calculate the acceleration of the car. Show clearly how you work out your final answer and give the units.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_

Acceleration = \_\_\_\_\_ (4)

- (d) The mass of the car is 900 kg.

- (i) Write down the equation that links acceleration, force and mass.

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(1)

- (ii) Calculate the force used to accelerate the car. Show clearly how you work out your final answer.

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Force = \_\_\_\_\_ newtons

(2)

(Total 11 marks)

**Q7.**

The stopping distance of a car is the sum of the thinking distance and the braking distance.

The table below shows how the thinking distance and braking distance vary with speed.

| Speed in m/s | Thinking distance in m | Braking distance in m |
|--------------|------------------------|-----------------------|
| 10           | 6                      | 6.0                   |
| 15           | 9                      | 13.5                  |
| 20           | 12                     | 24.0                  |
| 25           | 15                     | 37.5                  |
| 30           | 18                     | 54.0                  |

- (a) What is meant by the braking distance of a vehicle?

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(1)

- (b) The data in the table above refers to a car in good mechanical condition driven by an alert driver.

Explain why the stopping distance of the car increases if the driver is very tired.

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(2)

- (c) A student looks at the data in the table above and writes the following:

thinking distance  $\propto$  speed

braking distance  $\propto$  speed

Explain whether the student is correct.

(2)

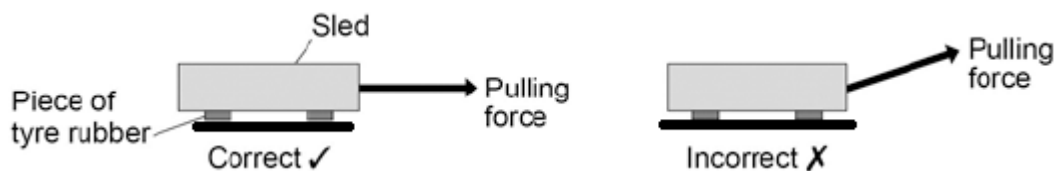
- (d) Applying the brakes with too much force can cause a car to skid.

The distance a car skids before stopping depends on the friction between the road surface and the car tyres and also the speed of the car.

Friction can be investigated by pulling a device called a 'sled' across a surface at constant speed.

The figure below shows a sled being pulled correctly and incorrectly across a surface.

The constant of friction for the surface is calculated from the value of the force pulling the sled and the weight of the sled.



Why is it important that the sled is pulled at a constant speed?

Tick **one** box.

If the sled accelerates it will be difficult to control.

☐

If the sled accelerates the value for the constant of friction will be wrong.

☐

If the sled accelerates the normal contact force will change.

☐

(1)

- (e) If the sled is pulled at an angle to the surface the value calculated for the constant of friction would not be appropriate.

Explain why.

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(2)

- (f) By measuring the length of the skid marks, an accident investigator determines that the distance a car travelled between the brakes being applied and stopping was 22 m.

The investigator used a sled to determine the friction. The investigator then calculated that the car decelerated at  $7.2 \text{ m/s}^2$ .

Calculate the speed of the car just before the brakes were applied.

Give your answer to two significant figures.

Use the correct equation from the Physics Equation Sheet.

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Speed = \_\_\_\_\_ m/s

(3)

(Total 11 marks)

## Mark schemes

### Q1.

- (a) Move the wooden block to the left. 1
- (b) use a pulley (on the edge of the bench)  
*allow any feasible method to stop the string from rubbing* 1
- (c) suitable scale 1
- points plotted correctly  
*allow 5 correctly plotted for 2 marks OR 3–4 correctly plotted for 1 mark* 2
- line of best fit 1
- (d) (directly) proportional  
*allow a correct description of direct proportionality*  
*ignore positive correlation*  
*allow weight (added to mass holder) for force*  
*allow  $f = ma$  for 1 mark* 1
- (e) repeat the measurements/investigation 1
- ignore anomalies **and** calculate the mean / average 1
- (f) resultant force = mass  $\times$  acceleration  
**or**  
 $F = m a$  1
- (g)  $0.375 = 0.60 \times a$  1
- $a = \frac{0.375}{0.60}$  1
- $a = 0.625 \text{ (m/s}^2\text{)}$  1
- $a = 0.63 \text{ (m/s}^2\text{)}$  1

[14]

**Q2.**

- (a) more streamlined

*accept decrease surface area*

1

air resistance is smaller (for same speed)

*accept drag for air resistance**friction is insufficient*

1

so reaches a higher speed (before resultant force is 0)

*ignore reference to mass*

1

- (b) (i) 1.7

*allow 1 mark for correct method, ie  $\frac{5}{3}$* *or allow 1 mark for an answer with more than 2 sig figs that rounds to 1.7**or allow 1 mark for an answer of 17*

2

- (ii) 7.5

*allow 1 mark for correct use of graph, eg  $\frac{1}{2} \times 5 \times 3$* 

2

- (iii) air (resistance)

*accept wind (resistance)**drag is insufficient**friction is insufficient*

1

**[8]****Q3.**

- (a) reference to

- weight / force of gravity / acting downwards

- unbalanced (by any upwards force)

*for 1 mark each*

2

- (b)
- ideas that*
- forces balance(d)

*gains 1 mark***but**

weight / force of gravity / downwards force balanced by friction / air resistance / drag / upwards force

*gains 2 marks*

latter increases with speed

(accept arrows or relevant length and direction if clearly labelled,

as answers to parts (a) and (b))  
for 1 further mark

3

[5]

#### Q4.

(a) air(resistance) has greatest effect on paper

1

(b) paper **or** both fall faster

1

(both) fall together

accept same speed **or** rate

1

[3]

#### Q5.

(a) (i) air resistance/drag/friction (or upthrust)  
weight/gravitational pull/gravity

for 1 mark each

1

(ii) air resistance/friction acts in opposite direction to motion

1

(iii) Y

1

(iv) the sky-diver accelerates/his speed increases  
in downward direction/towards the Earth/falls

for 1 mark each

2

(b) force X has increased force Y has stayed the same the speed of the sky-diver  
will stay the same

for 1 mark each

3

(c) (i) CD

1

(ii) 500  
(iii) 50 } (but apply e.c.f. from (i))

3

(iv) 10 (but apply e.c.f. from (ii) and (iii))  
gets 2 marks

or 500/50 or d/t

gets 1 mark

2

[14]

**Q6.**

- (a) concentration / tiredness / drugs / alcohol

*accept any reasonable factor that could affect a driver's reactions*

*do **not** accept speed or any physical condition unrelated to the driver*

1

- (b) 31.25

*credit for **1** mark correct attempt to calculate the area under the slope **or** for using the equation*

*distance = average velocity (speed) × time*

*credit for **1** mark use of correct velocity change (12.5) and correct time (5) **or** answer of 62.5*

3

- (c) 2.5

*credit for **1** mark triangle drawn on slope **or** correct equation **or** two correct pairs of coordinates*

*credit for **1** mark use of correct velocity change (12.5) and correct time (5)*

*accept time = between 4.8 and 5.2 if used in (b)*

*do not accept an attempt using one pair of coordinates taken from the slope*

3

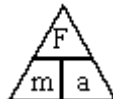
metres / second / second **or** metres / second / squared **or** m/s<sup>2</sup> **or** ms<sup>-2</sup>

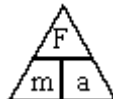
1

- (d) (i) force = mass × acceleration

*accept correct transformation*

*accept  $F = m \times a$*



*accept  provided subsequent use of Δ is correct*

*do **not** accept an equation in units*

1

- (ii) 2250

*credit their (c) × 900 for 2 marks*

*credit **1** mark for correct substitution*

2

[11]

**Q7.**

- (a) the distance travelled under the braking force

1

- (b) the reaction time will increase

1

increasing the thinking distance (and so increasing stopping distance)

*(increases stopping distance is insufficient)*

1

- (c) No, because although when the speed increases the thinking distance increases by the same factor the braking distance does not.

1

eg

increasing from 10 m / s to 20 m / s increases thinking distance from 6 m to 12 m but the braking distance increases from 6 m to 24 m

1

- (d) If the sled accelerates the value for the constant of friction will be wrong.

1

- (e) only a (the horizontal) component of the force would be pulling the sled forward

1

the vertical component of the force (effectively) lifts the sled reducing the force of the surface on the sled

1

- (f)  $-u^2 = 2 \times -7.2 \times 22$

*award this mark even with  $0^2$  and / or the negative sign missing*

1

$$u = 17.7(99)$$

1

18

1

*allow 18 with no working shown for 3 marks*

*allow 17.7(99) then incorrectly rounded to 17 for 2 marks*

[11]

## Examiner reports

### Q2.

- (a) Most students scored at least one mark, for referring to the more streamlined profile of design Y, and many went on to talk about the effect of air resistance. Nearly half of the students then referred their answer back to the effect on the top speed to gain full marks.
- (b)
  - (i) The majority of students were able to make the calculation accurately but many lost a mark by failing to give their answer to 2 significant figures. Many students incorrectly rounded down or put 1.6 recurring, denoted by a dot, not appreciating that this was not 2 significant figures.
  - (ii) This was poorly answered with the majority of the students scoring zero.
  - (iii) Less than half of the students answered this correctly.

### Q4.

This question was not well answered. Most answers showed a clear lack of understanding in part (a) and only a few candidates referred correctly to differences in air resistance. In (b) more candidates were successful but too many linked gravity with vacuum. Common answers included 'they will weigh the same in a vacuum' and 'there's no gravity in a vacuum'.

### Q5.

Almost all candidates gained marks for parts of this question, few gained all the marks available.

- (a) In (i) Y was attributed to weight/gravity by most candidates to gain a mark, far fewer were able to identify X as air resistance/drag/friction. A surprising number did come up with the non-syllabus answer of upthrust. A minority incorrectly identified the forces as a push and a pull. In part (ii) few could explain why X acted in an upward direction. In (iii) X was often given as the bigger force to gain the mark whilst in (iv) a clear statement that the sky-diver would accelerate downwards was far less common.
- (b) Many had force X increasing to gain the mark, fewer had force Y staying the same and fewer again the speed staying the same.
- (c) Very few candidates failed to gain some marks. Throughout this part errors were carried forward so that a mistake in an early part of the question was only penalised once. In (i) most correctly picked CD, in (ii) and (iii) most had 500 and 50 and went on in (iv) to a correct speed of 10m/s

### Q6.

- (a) Most candidates were able to gain the mark with alcohol/drugs or distractions. Those who failed to do so usually referred to weather/road conditions.
- (b) Candidates who read the graph correctly, and knew how to use the information obtained, were able to score maximum marks. Errors included misreading the time (5.2 or 6) or the final velocity (15). Of those candidates who used the equation, few gave distance = 'average' velocity (speed) × time.

- (c) Most candidates scored maximum marks using  $12.5/5 = 2.5$ , but not always showing clearly 'increase in velocity/time taken'. The majority of candidates gained the unit mark.
- (d) Most candidates were able to gain full marks provided that they had an answer in part (c).